

Statistical Learning

Lecture 02 Notes

1: Learning Definitions

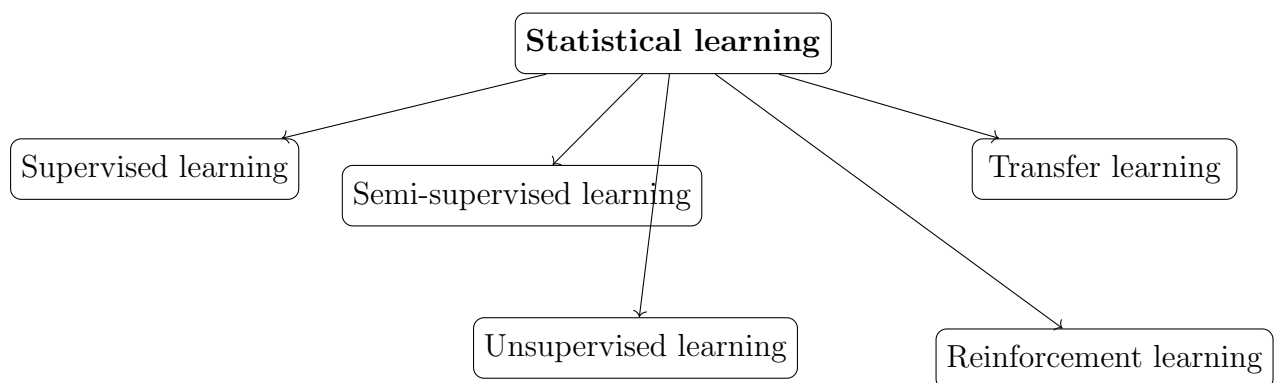
- **Learning** :- Extracting Knowledge / useful information from the data.
- **Mach. learning** :- We need to develop algorithms so that the machine Can do this job.
- **Statistical Mach. learning** :- We use statistical theories / methodologies to develop these algorithms.

2: What is Statistical Learning

Statistical Learning :-

It is branch of science where statistical theories and methodologies are used to develop algorithms for extracting meaningful information from the data.

3: Types of Statistical Learning (Flowchart)



4: Supervised Learning

Here we have a training set containing information on input (covariates) and output (response) variables and our aim is to study the relationship between the output and input variables so that we can predict the output for a new input as accurately as possible.

The output variable can be numerical or categorical.

- If we deal with a numerical response (output) it is called a **regression problem**.
- If the response is categorical, it is called a **classification problem**.

Formally, training set = $\{(\mathbf{X}_i, Y_i), i = 1, \dots, n\}$ where $\mathbf{X}_i \in \mathbb{R}^p$.

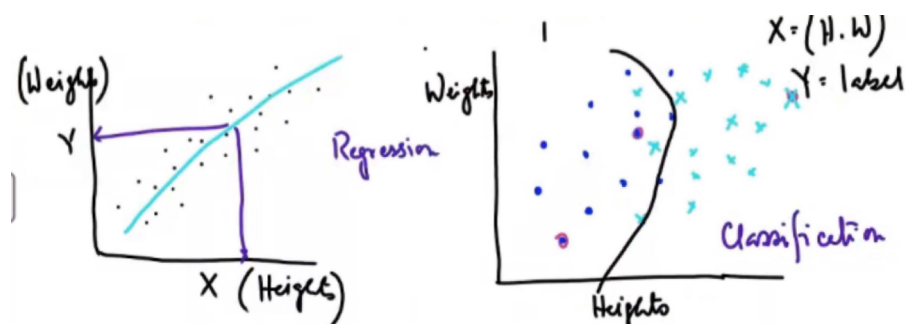
5: Unsupervised Learning

Here we do not have any labelled observation (response is not available) but still want to extract meaningful information from the data.

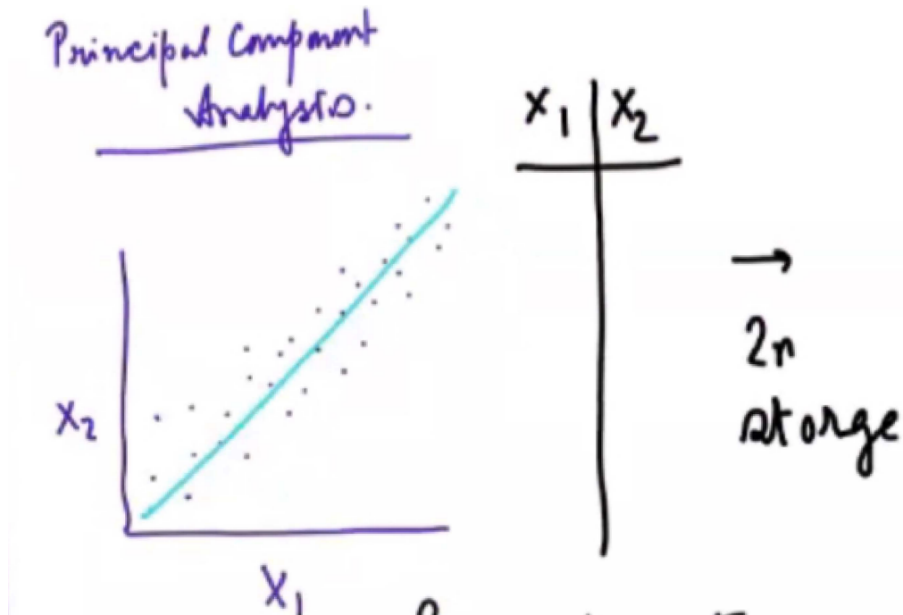
Common methods:

- Principal Component Analysis
- Multi dimensional Scaling
- Independent Component Analysis
- Cluster analysis

6: Regression vs Classification



7: Principal Component Analysis (PCA)



Assume $(0, 0)$ is the centre.

We look for direction vector $\alpha = (\alpha_1, \alpha_2)^T$ such that var of $\alpha^T \mathbf{x} = \alpha_1 X_1 + \alpha_2 X_2$ is maximized subject to

$$\|\alpha\| = \sqrt{\alpha_1^2 + \alpha_2^2} = 1.$$

Approximation:

$$\begin{pmatrix} x_{i1} \\ x_{i2} \end{pmatrix} \approx l_i \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

New storage: direction vector = 2, data points = $n \Rightarrow n + 2$

Also PCA can be viewed as finding lower dim representation that minimizes the error sum of squares (perpendicular residuals).

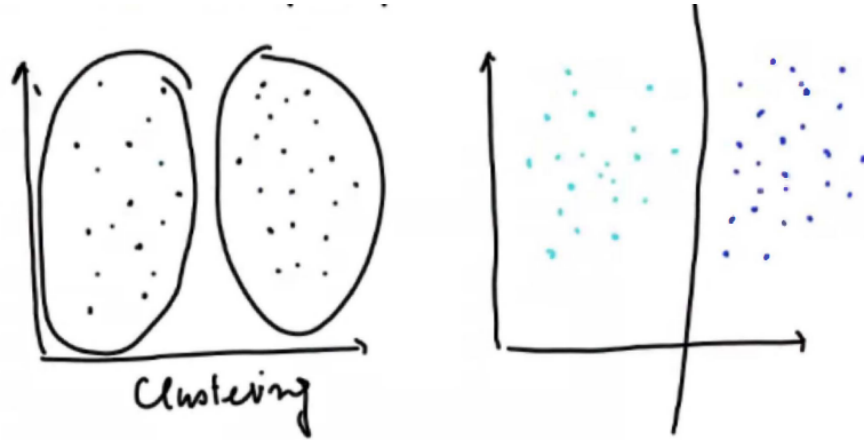
8: Other Methods

Multi-dimensional scaling :: It is also dimension reduction technique which considers linear/nonlinear projection of the observations such that the pairwise distances remains almost the same

Independent component Analysis (Blind source separation) :: It is method for extracting independent original signals from a data set containing mixtures of signals

9: Cluster Analysis

Cluster Analysis :: (unsupervised classification) Divide the data set into some homogeneous groups



Classification	Clustering
We have a training set of labelled observations	We do not have any labelled obsn.
Our aim to construct a discriminatory surface so that we classify the future (unlabelled) obsn as accurately as possible	Our aim is to divide this data set into several groups to understand the data better
The no. of classes are Known	No. of groups are unknown

10: Semi-supervised Learning



11: References and Evaluation

Ref:-

- Hastie, Tibshirani, Friedman : Elements of Stat. learning
- Duda, Hart, Stork : Pattern Classification.

Evaluation: 50 : Endsem, 30 : Midsem, 20 : Assign